



EDUCATION

Year	Degree/Exam	Institute	CGPA/Marks
2027	B.TECH	IIT Kharagpur	7.75 / 10
2022	CBSE	Santhome Higher Secondary School, Maihar	96%
2020	CBSE	Santhome Higher Secondary School, Maihar	94.2%

PROJECTS

- Real-Time Chat Application with Group Management | Self Project** [May'25-Jun'25]
- Developed a real-time web chat platform with one-to-one messaging, group chats, file sharing, for seamless cross-device communication
 - Implemented a real-time messaging system with <200ms latency, resolving fragmentation and supporting concurrent users in testing
 - Designed using **ReactJS**, **NodeJS**, **Express**, **MongoDB**, with **Socket.IO** for real-time messaging, **JWT** for auth, and **Multer** for file uploads
- Hotel Booking System with Real-Time Availability Management | Self Project** [Apr'25-May'25]
- Employed a hotel booking web app using the **MERN** stack with real-time availability, secure login, responsive UI, and dynamic filtering
 - Improved user experience by reducing booking steps by over **50%**, streamlining flow for users with better speed and accessibility
 - Used **ReactJS**, **NodeJS**, **Express**, **MongoDB**, and **RESTful APIs** for real-time updates, user authentication, and secure data transactions
- Radar Navigation and Height Detection | DIY Project** [Oct'23-Nov'23]
- Built a **radar-based system** using ultrasonic sensor and servomotor to find object distance and angle for navigation and height detection
 - Enabled real-time object height estimation using triangulation, simulating basic radar-based aerial tracking in constrained spaces
 - Utilized **Arduino Uno**, **ultrasonic sensor**, **servo motor**, and trigonometric algorithms for distance-angle computation and height detection
- Movie Recommender System Project | Self Project** [Feb'25-Mar'25]
- Designed a content-based movie recommendation app with an interactive UI, deployed using Heroku for seamless public access
 - Solved content overload by delivering highly personalized movie suggestions, significantly enhancing user discovery and engagement
 - Utilized **Python**, **Streamlit**, **scikit-learn**, **Pandas**, **NLTK**, and **Heroku** for efficient model building, data processing, and cloud deployment

COMPETITION/CONFERENCE

- Autonomous Maze Solving Competition | TeamKART, IIT Kharagpur** [Sep'23-Oct '23]
- Designed a versatile robotic system capable of autonomous maze-solving and Bluetooth-enabled racing using modular components for mobility, obstacle detection, and real-time control, adaptive logic to enable efficient navigation and dual-mode operation
 - Secured **Bronze** in autonomous navigation and **4th** place in manual race among **40+** teams by integrating real-time sensor data processing, algorithmic path planning, and efficient coordination between hardware components and control software for optimal performance
 - Utilized **Arduino IDE** for development, deploying programs on **ESP32 microcontrollers** with **ultrasonic sensors**, motor drivers, and Li-ion batteries, enabling manual control via Dabble App and implementing **Min-Max pathfinding algorithms** for autonomous navigation

SKILLS AND EXPERTISE

Programming Languages: Proficient: C | C++ | HTML | CSS | JavaScript Familiar: Python
Libraries/Frameworks/Tools: ReactJS | NodeJS | ExpressJS | MongoDB | Git | GitHub | MATLAB | C++ STL | VS Code | Arduino IDE | Microsoft Office | Socket.IO **Skills:** Data Structures And Algorithms | Competitive Programming | Web Development

AWARDS AND ACHIEVEMENTS

- Ranked in the **top 4% of 250,000 candidates** in **JEE Advanced 2023**, the qualifying exam for admission to the prestigious IITs across India
- Ranked in the **top 1% of 1.18 million candidates** in **JEE Main 2023**, a highly competitive national engineering entrance exam
- Cleared the **NDA Written Exam 2023** conducted by **UPSC**, ranking among the **top 1%** of candidates all over India
- Reached **Pupil** rank on Codeforces with a peak rating of **1267** through active participation in global rated rounds
- Secured an All India Rank (AIR) of 2969 in COMEDK UGET 2022, placing in the **top 4%** of candidates in this competitive national-level exam

COURSEWORK INFORMATION

Academics: Programming and Data Structures (Theory and Lab) | Linear Algebra, Numerical and Complex Analysis | Transform and Advanced Calculus | Chemical Engineering Thermodynamics | Heat transfer | Mass Transfer | Fluid Mechanics | Reaction Engineering | Chemical Process Calculation | Mechanical Operation **MOOCs:** Bootcamp on Python and Artificial Intelligence | Generative AI

POSITIONS OF RESPONSIBILITY

- Publicity And Marketing Subhead | Shaurya Sports Fest, IIT Kharagpur** [May'24 - Apr'25]
- Reached out to **500+** colleges with my team and successfully coordinated with participation of **20** colleges in Fest across India
 - Led a **7-member team** to manage accommodation logistics for visiting contingents, ensuring a smooth stay during the **3** days event
 - Organized and led meetings with fellow **Campus Ambassadors** to strategize outreach and enhance event publicity and participation
 - Managed **On-ground coordination** for Badminton matches, ensuring timely fixture execution and smooth flow of the event during Fest
- Secretary Alumni Affairs' | Patel Hall Of Residence** [Aug'24-Apr'25]
- Contributed to the **Media and Sponsorship team**; secured a barter deal for energy food packs to support blood donors during donation
 - Helped organize a hall-led **Orphanage Drive**; coordinated logistics and actively engaged with children during the outreach visit
 - Initiated and maintained a structured alumni database to streamline future communication, networking, and event coordination
 - Collaborated with hall administration and students to facilitate alumni interactions and support mentorship opportunities for residents

EXTRA CURRICULAR ACTIVITIES

Sports: **Silver Medalist** in Inter Hall Yoga Competition under General Championship, secured **fourth** place in Athletics relay race 4*400m
Cultural: **First position** in Zonal-level Skit Competition at school for outstanding scripting, direction, and team performance